

1. User Accounts and Groups in an Active Directory Domain

☒ User Accounts

- Represent individual users in the domain.
- Stored in the **Active Directory database (NTDS.dit)**.
- Created using tools like:
 - **Active Directory Users and Computers (ADUC)**
 - **PowerShell (New-ADUser)**
- Can be assigned passwords, group memberships, and permissions.

Types of User Accounts:

- **Domain User Accounts:** Managed centrally; can log into any domain-joined computer.
- **Local User Accounts:** Exist only on a local machine (not in AD).

☒ Groups

Groups help organize users and assign permissions efficiently.

Types of Groups:

- **Security Groups:** Used to assign rights and permissions.
- **Distribution Groups:** Used for email distribution lists (non-security).

Group Scopes:

- **Domain Local:** Used within the domain to assign permissions.
- **Global:** Can contain users from the same domain; used in trusted domains.
- **Universal:** Can contain users/groups from any domain in a forest.

Group Nesting Best Practice:

"AGDLP" – **A**ccounts go into **G**lobal groups, which are placed into **D**omain Local groups, which get assigned **P**ermissions.

2. Implementing an Active Directory Domain Services (ADDS)

☒ ADDS Overview

Active Directory Domain Services is the main role in Windows Server for creating and managing domain controllers. It provides:

- Centralized authentication (Kerberos)
- User and computer object management
- Group policies
- LDAP services

☒ Steps to Implement ADDS

1.

Install ADDS role using Server Manager or PowerShell:

powershell

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Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools

2.

Promote the server to a Domain Controller:

- Choose to add a new forest or domain.
- Provide domain name (e.g., corp.local).
- Configure Directory Services Restore Mode (DSRM) password.

3.

Reboot the system.

After installation:

- The server becomes a **Domain Controller**.

- Tools like ADUC, DNS, and Group Policy Management are available.

3. Domain Controller (DC)

A **Domain Controller** is a server that hosts a copy of the Active Directory database and handles:

- Authentication (login, Kerberos)
- Authorization (access to resources)
- Replication of AD data to other DCs

☒ Responsibilities:

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Stores NTDS.dit

- Hosts DNS services (commonly integrated)
- Maintains **FSMO roles** (Flexible Single Master Operations)

☒ **FSMO Roles:**

1.
Schema Master
2.
Domain Naming Master
3.
RID Master
4.
PDC Emulator
- 5.

4. Additional Domain Controller (ADC)

An Additional Domain Controller:

- Is a secondary DC in the same domain.
- Provides **redundancy** and **load balancing**.
- Holds a **replica** of the AD database.
- Participates in **multi-master replication**.

☒ **Benefits:**

- Fault tolerance
- Faster authentication
- Distributes login load across locations

Promotion:

You add an ADC by promoting a Windows Server to a domain controller **for an existing domain**.

powershell

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Install-ADDSDomainController -DomainName "corp.local"

5. Client (Windows 10/11)

A domain-joined client:

- Authenticates using domain credentials
- Applies Group Policy from the domain
- Accesses domain resources (shares, printers, etc.)

☒ Joining a Domain:

1. Go to System > About > Join a domain.
2. Provide domain name (e.g., corp.local).
3. Authenticate using a domain account.

4.

Restart the client.

After joining:

- The computer object is created in AD.
- Users can log in using domain credentials.

6. Member Server (MS)

A Member Server:

- Is part of the domain but **not** a domain controller.
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Does **not** store the AD database.

- Can host roles like File Server, Print Server, SQL Server, etc.

☒ **Uses:**

- Reduces the load on Domain Controllers.
- Ensures **separation of duties** (e.g., AD vs. file storage).
- Enables application hosting within domain security context.

☒ **Characteristics:**

- Appears in AD as a computer object.
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Receives Group Policies from the domain.

- Can use domain authentication for services.

Active Directory Domain Environment Structure

pgsql

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